

Experiment Pitching Rate and Oxygenation

From German brewing and more

Contents

- 1 Abstract
- 2 Introduction
- 3 Materials and Methods
- 4 Results and Discussion
 - 4.1 Fermentation performance
- 5 Taste test
- 6 Conclusion
- 7 Sources

Abstract

Esters are an important component of the aroma of German wheat beers. Common home brewing knowledge lists pitching rate and level of oxygenation as important factors that effect the level of esters that are produced during fermentation. While it is commonly believed among home brewers that lower pitching rates result in higher ester levels, the literature reports that increased pitching rates lead to higher levels of esters. This experiment is designed to evaluate the affect of oxygen levels and yeast pitching rate on the ester production.

While differences could be shown in the fermentation performance of the various fermentation samples, the taste and aroma of all the samples was remarkably similar which rendered this experiment inconclusive.

Introduction

Esters are formed through a condensation reaction between an alcohol and an acid [Wikipedia]. In brewing, 2 major processes exist in which esters are formed. During intra cellular ester formation, the yeast's metabolism produces esters through enzymatic reactions. But esters can also be formed by a simple condensation reaction between an organic acid and an alcohol. But due to the slow reaction rate, this form of esterformation doesn't play a role in primary fermentation and produces significant results only after extended aging (12+ weeks) [Hermann 2005, Narziss 2005]. This ester formation during aging is responsible for the dark fruit notes of aged beers. But more interesting for the brewing of Weissbier is the ester production during the primary fermentation.

During Fermentation pyruvic acid, an intermediate product of the pathway that leads to alcohol, is reduced to oxalacetate and then to acetyl CoA. Acetyl CoA is the basis for a host of compounds including sterols for cell wall construction and esters [Noonan 1996]. All authors agree that increased biomass production (creation of cell walls) reduces the Acetyl CoA that is available for ester production and leads to reduced ester levels in the beer [Narziss 2005, Clone, Walsh, Noonan 1996, Fix 1999]. Differences however exist with respect to ester production and yeast growth. Fix [Fix 1999] writes that

any increased activity on the acetyl CoA branch, i.e. yeast growth, will increase ester production while other authors [Narziss 2005, Clone] state that increased yeast growth leads to a decrease in esters since more of the acetyl CoA is used for sterol synthesis.

The following is a list of the factors that are known to affect the ester production:

- **Yeast strain:** the choice of yeast strain has a significant impact on the ester production. But according to Narziss is a yeast strains ability to form esters strongly dependent on the wort composition and high ester levels in one type of wort are not necessarily an indication that the same strain will also produce high levels of esters in another wort [Hermann 2005].
- **availability of oxygen:** It has been shown that increased oxygenation rates can lower the amount of esters produced by yeast [Narziss 2005, Noonan 1996, Fix 1999]. This has been explained by the increased production of biomass. When free oxygen is available, the acyl CoA is more likely to be used for the production of sterols and the amount available for ester production will be reduced.
- **Double batch (German: Drauflassen):** As noted earlier, any process that keeps the yeast in the growth phase decreased ester production. One of these methods is Drauflassen or double batch brewing. A fresh batch of aerated wort is added to the fermenter when the previous batch is at high krausen. This can be done multiple times and the continued yeast growth will reduce the amount acyl CoA that can be used for ester formation [Narziss 2005].
- **Temperature:** Higher fermentation temperatures are known to increase the production of fermentation byproducts including esters. Some authors have explained this with increased yeast metabolism. Saerens et. al. [Saerens 2007] gives this explanation "According to Suomaleinen [Suomalainen 1981], an increase in the fermentation temperature releases higher levels of esters through more efficient excretion and/or enhanced autolysis of the yeast. An effect of the temperature on the thermodynamic equilibrium of ester solubility in cellular lipids and the aqueous medium is another possibly more likely explanation". But it should be noted that the effect of temperature on the ester production is different between yeast strains and also vary between the types of esters that are produced [Hermann 2005]
- **pressure:** Even though increased CO₂ pressure leads to the retardation of yeast growth, it has been shown to reduce the amount of esters (and other fermentation byproducts) that are produced. The increased CO₂ pressure is affecting the synthesis of acetyl CoA which results in retarded yeast growth and lower ester production [Hermann 2005]. This shows that if yeast growth is slowed by reduced acetyl CoA levels, the ester production will not be increased.
- **pitching rate:** Higher yeast pitching rates lead to increased ester levels in the beer [Hermann 2005]. This statement conflicts with common home brewer knowledge [Fix 1999], that underpitching leads to less clean fermentations. According to Narziss, increased ester production is a result of the reduced yeast growth that results from high pitching rates [Narziss 2005].
- **FAN levels:** It has been shown that an increase in free amino nitrogen (FAN) leads to an increase in ester production [Saerens 2007]. Free amino nitrogen refers to the short length proteins that the yeast absorbs during fermentation. This leads to the conclusion that beers brewed with a high percentage of adjuncts (corn, rice) should show less ester production and that beers with a high FAN level (extended proteolytic activity during the mash. i.e. long protein rest at 50C) should contain more esters.
- **wort composition:** The higher the percentage of maltose in the wort, the less esters will be produced. Hermann showed that an increase in the worts glucose concentration leads to an increased ester production. The necessary increase in glucose can be achieved through corn-sugar

additions or mashing procedures that take advantage of the maltase enzyme. This enzyme can create glucose from maltose, but is not active in typical mashing as it is deactivated above 45 C [Hermann 2005]. Hermann explained that with a delayed occurrence of the diauxic point (when the yeast has to switch from glucose to maltose as their primary sugar source). This switch causes the yeast stress under which it produces more esters. If it is delayed, more yeast will encounter it due to the increased yeast growth that has happened up to this point [Hermann 2005].

- **wort gravity:** The higher the wort gravity, the more esters will be produced during fermentation [Saerens 2007]. The higher wort gravity provides more compounds for enzymatic reactions, the length of the fermentation is increased and the increased sugar concentration inhibits yeast growth which leads to more Acyl-CoA for ester production [Hermann 2005]

From a home brewers point of view only these parameters can be easily varied:

- yeast strain
- oxygenation
- pitching rate
- temperature
- wort composition (through mashing or sugar additions)

The objective of this experiment was to evaluate the qualitative effects of oxygenation and pitching rate on the aroma compounds created by a Weissbier yeast.

Materials and Methods

For this experiment a 12 Plato wort was brewed from the following grist:

- 70% Weyermann light Wheat
- 30% Weyermann Bohemian Pilsner

A Hochkurz mash with the following rests was used to mash the grain:

- Dough-in, 59 C, 15 min, infusion
- Maltose, 63 C, 45 min, infusion
- Dextrinization, 69 C 20 min, infusion
- Mash-out, 76 C, 10 min, decoction

The 25L preboil volume wort was then boiled for 60 min with 10g Target hops (10% a-acids). The wort was cooled to a pitching temperature of 17C. 9 2 liter sanitized PET Pepsi Cola bottles were filled with 1.5 l each. The wort was then oxygenated using a sintered stainless steel oxygen stone and a regulator. Previously it was determined that the flow rate of oxygen from the oxygen stone at the chosen regulator setting is about 0.6 mg/s. After that the yeast was pitched. The used yeast originated from a pack of Wyeast 3056. I later realized that this is actually a blend and since I have been storing and propagating this yeast for a while I have no idea how the relationship between the yeast strains that were blended changed. The pitched yeast culture came from the slurry of a primary fermentation that started 2 weeks earlier.

The fermentation samples were oxygenated and pitched as follows:

A	B	C
1.6 ml/l yeast slurry; 1 s oxygen (0.4 ppm)	5.0 ml/l yeast slurry; 1 s oxygen (0.4 ppm)	15 ml/l yeast slurry; 1 s oxygen (0.4 ppm)
D	E	F
1.6 ml/l yeast slurry; 3 s oxygen (1.2 ppm)	5.0 ml/l yeast slurry; 3 s oxygen (1.2 ppm)	15 ml/l yeast slurry; 3 s oxygen (1.2 ppm)
G	I	H
1.6 ml/l yeast slurry; 9 s oxygen (3.6 ppm)	5.0 ml/l yeast slurry; 9 s oxygen (3.6 ppm)	15 ml/l yeast slurry; 9 s oxygen (3.6 ppm)

The oxygenation times were taken from my standard ale oxygenation time of 60s for 20l of wort. I later calculated the amount of O₂ that was released from the stone, but not all of the released O₂ was actually absorbed by the wort. In addition to that the head space also contained oxygen, even more than was added through the oxygen stone. As a result of that the oxygenation part of this experiment cannot be deemed reliable.

The pitching rates amount to the following pitching rates for 20l wort:

- 1.6 ml/l -> 32 ml yeast slurry
- 5 ml/l -> 100 ml yeast slurry
- 15 ml/l -> 300 ml yeast slurry

The actual number of viable cells was not determined

The fast ferment test showed a limit of attenuation of 78% (apparent extract = 2.6 Plato).

All samples were fermented in a single water bath at an average temperature of 19 C (peaked at 20C on day 4). On day 6 the samples were bottled in 12 oz bottles with the addition of corn sugar for carbonation. The amount of corn sugar was based on the current extract level of the beer in order to achieve similar carbonation levels despite the fact that the samples had different extract levels at the time of bottling.

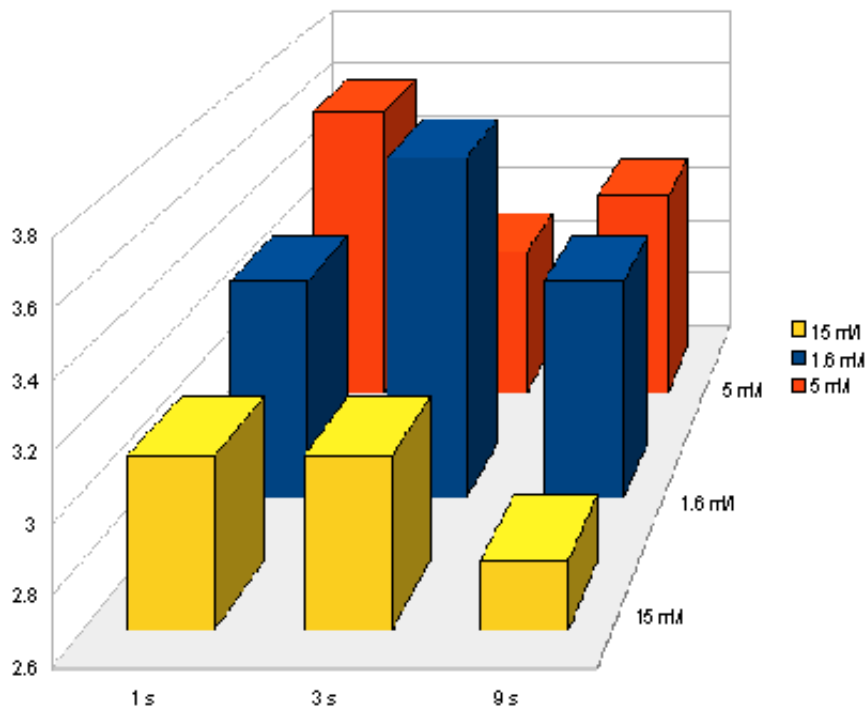
The bottles were then stored at about 19C for 8 more days before they were moved to an ambient temperature of 15 C where they sat before sampling.

Results and Discussion

Fermentation performance

On day 5 the following extract levels were measured for the 9 samples:

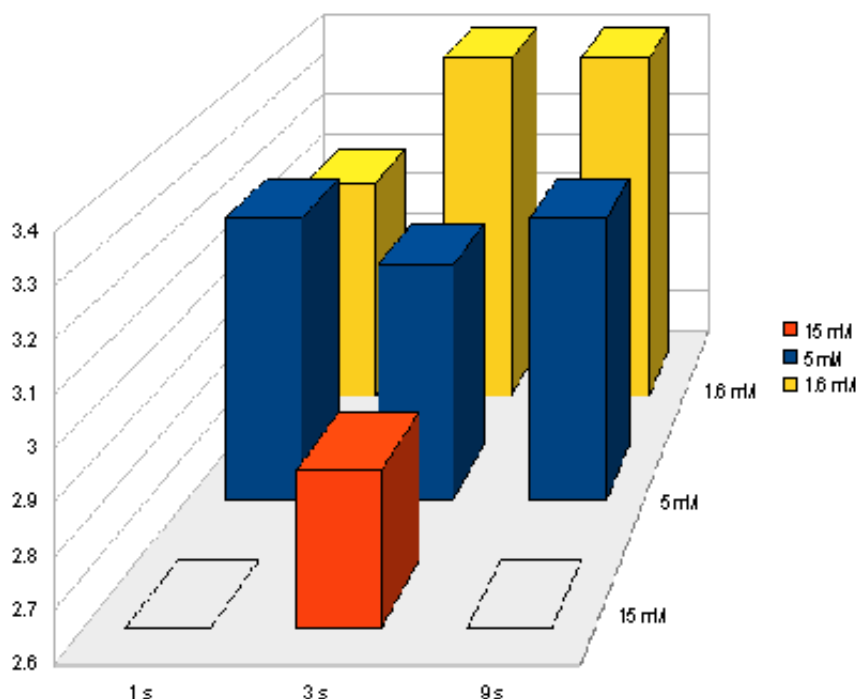
oxygen time	pitching rate		
	1.6 ml/l	5 ml/l	15 ml/l
1 s	3.3	3.6	3.1
3 s	3.7	3.1	3.1
9 s	3.3	3.3	2.8



With some outliers, the results matches the expectation that higher pitching rates and more oxygen lead to a faster fermentation. The floor in the graph has been set to the extract level that was determined though the fast ferment test (2.6 Plato).

On day 6 (bottling day), the extract levels were not as consistent

oxygen time	pitching rate		
	1.6 ml/l	5 ml/l	15 ml/l
1 s	3.1	3.2	2.6
3 s	3.4	3.1	2.9
9 s	3.4	3.2	2.6



While the samples don't seem to follow the expected trend for oxygenation times, they seem to follow the expected trend for the pitching rates. Two of the samples with the highest pitching rate were able to reach the limit of attenuation (78% or AE = 2.6 Plato). The absence of a clear relationship between the oxygenation rate and the fermentation performance is assumed to be a result of the imprecise oxygenation which was discussed earlier.

Taste test

The following are notes on the aroma and taste tests that were done on the various fermentation samples. All taste tests were done by only one taster and none of the beers exhibited a strong ester profile to begin with. This was an indication that the used yeast may not have been the best for this experiment.

12 days after bottling the samples A, B and C (lowest oxygenation, 3 different pitching rates) were evaluated for taste and aroma. It was noted that sample A was less estery than sample C but the difference was very subtle. Actually so subtle that three triangle blind tests between samples A and C, that were done in succession, did not yield a conclusive result. A slight "band aid" taste and aroma was also noted for sample A, but not strong enough for sample A to stand out in the triangle test.

18 days after bottling, the samples G, H and I (highest oxygenation, 3 different pitching rates) were evaluated for taste and aroma. The results were similar to the first taste test and no conclusive difference was found in 3 triangle tests between G and I, though it was noted that H showed slight "band aid" aroma and taste immediately after pouring.

20 days after bottling D, E and F (medium oxygen, 3 different pitching rates) were evaluated. Sample D (lowest pitching rate) showed a band aid aroma and taste, but besides this the aroma and taste between all three samples is very close and it was difficult to tell them apart.

21 days after bottling B and H (medium pitching rate, lowest and highest oxygen) were compared. No "band aid" was found in either sample but B was regarded as less clean and appeared to have more fusel alcohols. But the taste and aroma of both samples was almost the same. No difference in the ester levels was perceived.

31 days after bottling A (lowest oxygen and lowest pitching rate) was compared against I (highest pitching rate and oxygen). Both were remarkably close to each other but I seemed more "rounded" in its taste than A without being able to say exactly why. No significant "band aid" was found in either sample.

Given the fact that none of the performed blind tastings was able to definitively tell a difference between 2 beers, it is certainly possible that the little differences that were perceived were influenced by the knowledge of which sample was tasted.

Conclusion

Despite the 9 fold difference in pitching rate and oxygen injection, which showed differences in the fermentation performance, the samples tasted remarkably similar and none of the blind taste tests could tell them apart. The actual difference in dissolved oxygen that was available to the yeast was likely much smaller, but the pitching rates were definitely widely different.

This experiment needs to be seen as inconclusive and further experiments need to be done with a different yeast. But at that point only 3 or 4 fermentations should be run to test either the influences of the pitching rate or the oxygen levels. Having only 3 to 4 samples that need to be evaluated is more manageable than 9 samples.

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